

# Collin Aldaibis

AI ENGINEER

AI engineer with an MSc in Computing Science. I build production AI systems end to end, from training and fine-tuning language models to the web apps and infrastructure that ship them, and enjoy working directly with stakeholders to turn business goals into working software.

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## ■ EXPERIENCE

### ○ Machine Learning Engineer at Netherlands Forensic Institute (NFI) 2024 – Aug 2026

The Hague, Netherlands

Building forensic AI in varying R&D projects, leading the engineering and LLM work in collaboration with the police, the Hansken team, and the biological and chemical data labs.

- Fine-tuned Whisper for Dutch forensic audio (ethnolects, dialects, and low-quality recordings), cutting word error rate from **~40%** to **~15%** on private forensic datasets and teaching it previously unseen slang.
- Initiated and supervised a master's thesis on fine-tuning Whisper for new vocabulary using text-only data, without requiring corresponding audio recordings.
- Pre-trained a Dutch ModernBERT from scratch on Dutch FineWeb / FineWeb-Edu to close a gap in open-source Dutch encoder models with a modern BERT/RoBERTa successor.
- Built and evaluated the Hansken Copilot RAG application, helping investigators search cases of millions of digital traces, fully on-prem with local embedding models and LLMs, evaluated with RAGAS on hand-curated forensic datasets.
- Benchmarked LLMs (GPT-NL, Mistral, Gemma, Qwen) with SURF & TNO; contributed the first Dutch bias benchmark to EuroEval and built a fine-tuning/eval pipeline for a forensic text-classification use case to assess whether GPT-NL could outperform existing models.
- Established local LLM infrastructure for running and fine-tuning models on sensitive data (llama.cpp, Unsloth, Hugging Face, PyTorch), including sandboxed local coding agents and optimised training runs on NVIDIA H200s.
- Founded and chaired the Python Dev Club and LLM Club to raise engineering and LLM literacy across the guild (modern tooling like Ruff and uv, plus safe local agents), and advised NFI leadership on rolling out LLM technology organisation-wide.

Python · PyTorch · HF datasets, TRL, transformers · Unsloth · llama.cpp · Streamlit

### ○ Full-Stack Software Engineer at Cordys Analytics 2023 – 2024

Utrecht, Netherlands

Built the company's core AI product, giving clinicians and patients seamless access to AI-powered ECG diagnostics.

- Developed the diagnostics platform end to end, from model serving to clinician-facing UI.
- Supervised several international AI master's students on deep learning for ECG research theses.
- Organised company events and team activities.

○

Software & Machine Learning Engineer at Schiphol (via Incentro)

2022 – 2023

Amsterdam, Netherlands

Delivered ML and full-stack applications for Schiphol as a consultant, owning the relationship with the customer.

- Built and maintained a Flask ML service, a Laravel backend, and an Angular frontend.
- Worked directly with stakeholders at Schiphol to translate business objectives into delivered software.
- Applied NLP techniques, word embeddings, and LLMs to production use cases.

Python · Flask · Laravel (PHP) · Angular (TypeScript) · GCP · Docker · MySQL · CI/CD

## PERSONAL PROJECTS

### Aparecium: Agentic LLM Benchmark Extraction ↗

A multi-stage agentic pipeline that extracts AI model benchmark scores from official model pages, model cards, and benchmark charts: web search, content/table parsing, OCR on chart images, LLM-based extraction, and resolution against a canonical benchmark catalog, evaluated against ground truth. Includes a self-improving agent that diagnoses recurring failure modes, applies targeted fixes, and re-validates, plus a frontend for exploring and comparing results.

Python · LLM agents · OCR · TypeScript · uv

### Speechless: Whisper Vocabulary Adaptation ↗

Exploratory research on adapting Whisper to out-of-vocabulary terms (e.g. Dutch medical vocabulary) without training on real audio: a single-codebook residual VQ over frozen Whisper encoder states and a GPT-2-style decoder mapping text to RVQ token sequences. Reproduces and extends the Speechless (2025) methodology.

Python · PyTorch · Whisper · ASR

### Fully Autonomous Agent for Rental Applications ↗

End-to-end agentic automation: triggered by incoming email, it opens a target listing, extracts metadata, and hands off to a browser agent that navigates third-party sites, writes a tailored message from a template, fills the application form, uploads documents, and submits, fully autonomously.

Python · Langchain Deep Agents · Playwright

## EDUCATION

Utrecht University

2018 – 2022

MSc, Computing Science

Master's thesis "Investigating relaxed probability updating games" (grade 8.2). Coursework: advanced machine learning, data mining, pattern recognition, evolutionary computing, multi-agent learning, network science.

Utrecht University

2015 – 2018

BSc, Computing Science

Coursework: algorithmics, optimisation, concurrency, security, graphics, computational intelligence, functional programming.

## CERTIFICATIONS

AWS Certified Machine Learning (Specialty) · Amazon Web Services

2023

Associate Cloud Engineer · Google Cloud

2023

Deep Learning Specialisation · DeepLearning.AI (Coursera)

2022